

Solution Requirements

Date	29 September 2026
Team ID	SWTID- 2026- 3665
Project Name	EduGenie – AI Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non- Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

Functional Requirements (FR) define specific behaviors, functions, or features of the system. Non- Functional Requirements (NFR) specify criteria that judge how the system should operate.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	User can access the EduGenie learning assistant through the application interface; user input and session data should be handled securely.	High
2	Authorization levels	Student users can access learning features such as notes, quizzes, explanations and study support. Administrative access, if added, is restricted to authorized users.	Medium
3	External interfaces	Integrates with the AI model/API used by EduGenie to generate explanations, summaries, quizzes and learning assistance.	High
4	Transactions processing	Accepts a learning request, sends the required prompt/data to the AI service, receives the generated result, and displays it to the user.	High
5	Reporting	Provides generated notes, quiz results, explanations and learning- progress information in a readable format.	Medium
6	Business rules, etc.	The system should generate learning content from the user's topic or question and provide relevant explanations, quizzes and study support.	High
7	Compliance to laws or regulations	Protect user data, avoid exposing API keys or sensitive configuration, and follow applicable data- privacy and responsible- AI requirements.	High
8	Other	Provide an easy- to- use interface for learning assistance, including notes/ summaries, quizzes, question answering and personalized learning support.	High

Step 2: Non- Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance&Speed	The interfaces should respond promptly and AI-generated results should be displayed without unnecessary delay.	Normal page interactions should respond within 2 seconds; AI response time depends on the external AI service.
2	Scalability	The application should support growth in users, learning requests and stored learning data.	Design should allow additional users and requests without major redesign.
3	Security&Data Privacy	Protect user inputs, credentials and configuration. Keep API keys and secrets out of source code and public repositories.	Secrets stored securely; no API key exposed in client-side code or repository
4	Reliability&Availability	The application should handle temporary service/API failures gracefully and provide a useful error message.	Graceful error handling for failed AI requests; application remains usable for supported local features.
5	Usability&Accessibility	The interfaces should be simple, readable and easy for students to navigate.	A first-time student should be able to enter a topic/question and access the generated learning support without assistance.
6	Other	Maintainable and portable code structure with clear modules for learning features and future enhancements.	Code organized into reusable modules; easy to update or extend with additional learning features.

Team Members – Group 2

No.	Member Name	Group No.
1	MADHUMIDHA V	2
2	Ragavi S	2
3	Bharani C	2
4	Vinayashree S	2